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MINERALS AND ENERGY RESOURCES

Geography - Class 10 | Chapter 5 | Deep Structured NCERT Notes

1. Introduction to Minerals

What are Minerals?

Minerals are homogeneous, naturally occurring substances with definite chemical composition and crystalline structure. Minerals are **found in nature** and are **formed through geological processes**. Minerals are the **building blocks of rocks**.

Minerals are essential for human civilization. They are used for **construction, manufacturing, electronics, jewelry, and energy production**. **Almost everything we use contains minerals** in some form.

Minerals are non-renewable resources meaning they **take millions of years to form**. Once extracted and

used, they **cannot be replaced** within human time scale. **Conservation of minerals is essential** for **future generations**.

How Minerals are Formed

Minerals are formed through various geological processes over **millions of years**. **Magmatic processes** form minerals when **molten magma cools and solidifies**. **Metamorphic processes** form minerals when **existing rocks are changed by heat and pressure**.

Sedimentary processes form minerals when **materials are deposited and compressed** over time.

Hydrothermal processes form minerals when **hot water solutions deposit minerals** in cracks and cavities. **Weathering and erosion** can also concentrate minerals.

Minerals occur in different forms:

- **In igneous and metamorphic rocks as veins and lodes**
- **In sedimentary rocks as beds or layers**
- **As alluvial deposits in valley floors and hillsides**
- **As ocean water dissolved minerals**

2. Distribution of Minerals in India

Mineral Resources of India

India is rich in mineral resources but **not self-sufficient in all minerals**. India has **abundant reserves of iron ore, mica, bauxite, and manganese**. However, India is **dependent on imports for several minerals** like **copper, lead, zinc, gold, and diamonds**.

Mineral distribution in India is uneven. Peninsular plateau region has most of India's mineral reserves. Western and eastern coasts have concentrated deposits of certain minerals. Northern and north-eastern regions have limited mineral resources.

India's mineral wealth is concentrated in a few states:

- **Jharkhand, Odisha, Chhattisgarh:** Iron ore, coal, manganese
- **Rajasthan:** Zinc, lead, copper, building stones
- **Andhra Pradesh, Odisha:** Bauxite
- **Karnataka, Jharkhand:** Iron ore
- **Gujarat, Rajasthan:** Petroleum

3. Types of Minerals

Ferrous Minerals

Ferrous minerals contain iron and are **magnetic in nature**. Ferrous minerals are **important for steel industry**. India has **abundant reserves of ferrous minerals**.

Iron Ore:

- **Iron ore is the basis of industrial development.**
Steel is made from iron ore. India is one of the important producers of iron ore in the world.
- **Two main varieties: Magnetite** with high iron content and **Hematite**, the most important industrial iron ore.
- **Magnetite** is the **most valuable iron ore** due to **high iron content** and **magnetic properties**.
- **Hematite** is the **most widely used iron ore** because it is found in large quantities and is suitable for steel-making.

Major iron ore belts:

- **Odisha-Jharkhand belt:** Badampahar in Odisha and Gua in Jharkhand.
- **Durg-Bastar-Chandrapur belt:** Chhattisgarh and Maharashtra; includes **Bailadila mines**.

- **Bellary-Chitradurga-Chikkamagaluru-Tumakuru belt:** Karnataka; includes **Kudremukh mines**.
- **Maharashtra-Goa belt:** Goa and Ratnagiri in Maharashtra.

Manganese:

- **Manganese is primarily used for making steel and ferro-manganese alloys.**
- **Manganese is also used in bleaching powder, insecticides, and batteries.**
- **Major producing states: Odisha, Karnataka, Maharashtra, Madhya Pradesh, and Andhra Pradesh.**

Nickel, Chromium, and Cobalt are also **ferrous minerals used in alloy making**. India has **limited reserves** of some alloy minerals and depends on imports in several cases.

Non-Ferrous Minerals

Non-ferrous minerals do not contain iron and are **non-magnetic**. They are **important for electrical, electronics, aircraft, transport, and communication industries**. India has **limited reserves** of several non-ferrous minerals.

Bauxite:

- **Bauxite is the ore of aluminum.** Aluminum is **light, strong, and corrosion-resistant.**
- **Bauxite is formed by decomposition of rocks** rich in aluminum silicates.
- **Aluminum is used in aircraft, automobiles, packaging, and construction.**
- **Major producing states: Odisha, Gujarat, Jharkhand, Maharashtra, Chhattisgarh, Madhya Pradesh, Tamil Nadu, and Kerala.**
- **Amarkantak plateau** is an important bauxite-producing region.

Mica:

- **Mica is a mineral made up of a series of plates or leaves.**
- **Mica splits easily into thin sheets** that can be **transparent.**
- **Mica has excellent electrical properties** and is **non-conductive.**
- **Mica is used in electrical and electronic industries** for **insulation.**

- **Major producing states: Jharkhand, Andhra Pradesh, Rajasthan, Bihar, Maharashtra, Tamil Nadu, and Kerala.**

Copper:

- **Copper is malleable, ductile, and an excellent conductor of electricity and heat.**
- **Copper is used in electrical cables, electronics, and chemical industries.**
- **India has limited reserves and is dependent on imports.**
- **Major producing states: Madhya Pradesh, Rajasthan, Jharkhand, Andhra Pradesh, Karnataka, and Telangana.**
- **Khetri copper belt** in Rajasthan is important.

Lead and Zinc:

- **Lead is used in batteries, paints, and radiators.**
- **Zinc is used for galvanizing iron and making alloys.**
- **Major producing states: Rajasthan, Andhra Pradesh, Jharkhand, Bihar, and Gujarat.**

Gold:

- **Gold is a precious metal used for jewelry and investment.**
- **India has limited reserves and is heavily dependent on imports.**
- **Major producing areas: Karnataka, Jharkhand, Andhra Pradesh, and Tamil Nadu.**

Non-Metallic Minerals

Non-metallic minerals do not contain metals. They are **used in construction, chemicals, and manufacturing.** India has **adequate reserves of several non-metallic minerals.**

Limestone:

- **Limestone is made of calcium carbonate.**
- **Limestone is the basic raw material for cement industry.**
- **Limestone is also used in steel industry as flux.**
- **Major producing states: Andhra Pradesh, Gujarat, Rajasthan, Madhya Pradesh, Tamil Nadu, Chhattisgarh, Maharashtra, Haryana, Karnataka, and Odisha.**

Rock Minerals:

- **Marble** is a **metamorphic rock** used for **sculpturing and construction**; **Rajasthan** is the largest producer.
- **Slate** is used for **roofing and writing boards**; **Rajasthan** is an important producer.
- **Granite** is used for **construction and decoration**; important states include **Andhra Pradesh, Karnataka, Tamil Nadu, Rajasthan, and Kerala**.

4. Conservation of Minerals

Why Minerals Need Conservation

Minerals are non-renewable resources that **take millions of years to form**. The **current rate of extraction is much faster than the formation rate**. Once minerals are depleted, they **cannot be replaced within human time scale**.

Mineral reserves are finite and decreasing. **High demand for minerals** due to **industrialization and urbanization** increases pressure on reserves. **Overexploitation leads to depletion of minerals**.

Conservation of minerals is essential for:

- **Sustainable development**

- **Economic security** because industries depend on minerals
- **Environmental protection** because mining causes damage
- **Future generations** because minerals must be preserved

Methods of Mineral Conservation

1. Planned and Sustainable Extraction: Minerals should be extracted in a planned manner with sustainable practices. Overexploitation should be avoided. Mining should follow **scientific and environmental guidelines**.

2. Use of Improved Technology: Improved technology can make low-grade ores economically viable. Advanced extraction and processing techniques reduce waste. **Technology improves efficiency** and **reduces environmental impact**.

3. Recycling of Metals: Recycling metals reduces demand for virgin ores. Scrap metal can be melted and reused. Steel, aluminum, and copper can be recycled efficiently.

4. Use of Substitutes: When possible, substitute materials should be used for scarce minerals. **Alternative materials** reduce pressure on rare minerals.

Scientific research can create **new substitutes**.

5. Reducing Waste: Waste during mining, processing, and manufacturing must be reduced.

Better recovery from mines helps use resources fully.

Efficient use of minerals increases their life.

6. Developing Sustainable Alternatives: Renewable energy reduces dependence on fossil fuels. **New materials** can replace scarce minerals. Research and innovation are necessary for long-term resource security.

5. Energy Resources

Types of Energy Resources

Energy resources are classified into conventional and non-conventional sources. Energy is needed for **agriculture, industry, transport, domestic use, and communication.** Without energy, modern economic life cannot function.

Conventional Energy Sources:

- **Traditional sources** used for a long time.
- **Mostly non-renewable** except firewood.
- **Include** firewood, cattle dung, coal, petroleum, natural gas, and electricity from thermal and hydel sources.

- **Cause environmental pollution** in many cases.
- **Finite and depleting.**

Non-Conventional Energy Sources:

- **Modern sources** developed more recently.
- **Mostly renewable and sustainable.**
- **Include** solar, wind, tidal, geothermal, biogas, and nuclear energy.
- **Environmentally friendly** compared to fossil fuels.
- **Inexhaustible** in many cases.

Conventional Energy Resources - Coal

Coal is the most abundant fossil fuel in India. Coal is **formed from plant remains** that were **buried and compressed over millions of years.** Coal is **used for electricity generation and industrial purposes.**

Coal is India's most important energy source:

- **Used in thermal power plants for electricity generation**
- **Used in steel, cement, and other industries**
- **Important for industrial development**

- **India is one of the major coal producers in the world**

Varieties of Coal:

- **Anthracite: Highest quality coal**, hard, dense, and burns with little smoke.
- **Bituminous: Most popular commercial coal**, high calorific value and widely used.
- **Lignite: Low-grade coal**, high moisture content and found in **Neyveli, Tamil Nadu**.
- **Peat: Lowest quality coal**, first stage of coal formation and not commercially important.

Major Coal Fields in India:

- **Gondwana Coal:** Damodar Valley, Jharia, Raniganj, Bokaro, Talcher, Korba, and Singareni.
- **Tertiary Coal:** Neyveli, Assam, Rajasthan, Jammu and Kashmir, and Arunachal Pradesh.
- **Major mines: Jharia, Raniganj, Bokaro, Talcher, Korba, and Singareni.**

Conventional Energy Resources - Petroleum

Petroleum is the second most important energy source in India. Petroleum is **formed from organic matter** that was **buried and subjected to heat and pressure**. Petroleum is called **black gold** due to its **economic importance**.

Petroleum provides:

- **Fuel for transportation** like petrol and diesel
- **Raw material for petrochemical industries**
- **Lubricants for machinery**
- **Important energy support for industry and domestic life**

India is dependent on imports for petroleum.

Petroleum imports create **major expenditure** for India. Therefore, **conservation and alternative energy sources** are important.

Petroleum Occurrence:

- **Petroleum occurs in anticlines and fault traps** in **sedimentary rocks**.
- **Oil is trapped in porous rock layers** covered by **impermeable rock**.

- **Petroleum is often found with natural gas.**

Major Petroleum Fields:

- **Mumbai High:** largest offshore petroleum field, located off the coast of Maharashtra.
- **Ankleshwar:** important onshore field in Gujarat.
- **Assam:** includes **Digboi, Naharkatiya, Moran,** and **Rudrasagar.**
- **Krishna-Godavari Basin:** important newer exploration region.
- **Kaveri Basin:** important basin in southern India.
- **Gujarat:** includes **Ankleshwar, Lunez,** and **Kalol.**

Conventional Energy Resources - Natural Gas

Natural gas is found in association with petroleum.

Natural gas is a **cleaner fuel** than coal and petroleum. It is used for **electricity generation** and as a **raw material for petrochemical industries.**

Natural Gas Properties and Uses:

- **Environment-friendly fuel** with less pollution.
- **Used in power generation and fertilizer industry.**
- **Compressed Natural Gas (CNG)** is used for vehicles.

- **Liquefied Natural Gas (LNG)** is used for transportation and storage.

Major Natural Gas Fields:

- **Krishna-Godavari basin**
- **Mumbai High**
- **Assam**
- **Gujarat**
- **Tripura, Rajasthan, and Jammu and Kashmir**

Hazira-Vijaipur-Jagdishpur pipeline is an important gas pipeline. **Gas Authority of India Limited (GAIL)** handles gas distribution.

Electricity

Electricity is generated in two main ways: hydroelectric power and thermal power. Electricity is essential for **industries, homes, transport, communication, irrigation, and services.**

Hydroelectric Power:

- **Generated by flowing water turning turbines.**
- **Renewable energy source.**
- **Major hydro projects** include **Bhakra-Nangal, Damodar Valley, Mahanadi, Krishna, and Godavari**

projects.

- **Important states** include **Himachal Pradesh, Karnataka, Tamil Nadu, Maharashtra, and Uttar Pradesh.**

Thermal Power:

- **Generated by burning coal, petroleum, or natural gas.**
- **Non-renewable and causes pollution.**
- **Major thermal power centres** include **Badarpur, Korba, Talcher, and Tuticorin.**

Non-Conventional Energy Resources

Solar Energy is obtained from the **sun**. It is **renewable and clean**. India has **abundant solar potential** due to its **tropical location**. Solar energy is used in **solar panels, cookers, water heaters, street lights, and power plants.**

Wind Energy is obtained from **wind turning turbines**. It is **renewable and clean**. Important wind energy states include **Tamil Nadu, Gujarat, Karnataka, Maharashtra, Andhra Pradesh, Rajasthan, and Kerala.**

Tidal Energy is obtained from **ocean tides**. It is **renewable and predictable** but has **limited suitable sites**. Important areas include the **Gulf of Kutch, Gulf of Khambhat**, and **Cochin backwaters**.

Geothermal Energy is obtained from **heat within the Earth**. Important areas include **Manikaran, Puga Valley, Tatapani**, and **Godavari basin**. It is **renewable, reliable, and clean**.

Biogas is obtained from **decomposition of organic matter** like **plant waste, animal waste, and human waste**. It is also called **gobar gas**. It is used for **cooking, lighting, electricity generation**, and produces **organic manure** as a by-product.

Nuclear Energy is obtained from **nuclear reactions** using **uranium and thorium**. It has **high energy output** and **low emissions**, but also has **waste and safety concerns**. Important nuclear power stations include **Rawatbhata, Kalpakkam, Tarapur, Narora, Kaiga, Kakrapar**, and **Kudankulam**.

6. Conservation of Energy Resources

Why Energy Conservation is Important

Energy resources are finite and depleting. The current consumption rate is unsustainable. Conservation is essential for sustainable development.

Energy conservation benefits:

- **Saves money** on energy bills
- **Reduces environmental pollution**
- **Conserves resources for future generations**
- **Reduces dependence on imports**
- **Promotes energy security**

Methods of Energy Conservation

1. Use Public Transport: Public transport reduces per-person energy consumption. Carpooling saves fuel. Walking and cycling save energy for short distances.

2. Switch Off When Not in Use: Switch off lights, fans, and appliances when not needed. This simple habit saves a large amount of electricity. It also reduces power bills.

3. Use Energy-Efficient Appliances: **BIS-certified appliances** consume less energy. **LED bulbs** save energy compared to incandescent bulbs. Efficient appliances reduce long-term energy demand.

4. Use Renewable Energy: **Solar panels, wind turbines, and biogas plants** reduce dependence on fossil fuels. Renewable energy is cleaner. It supports sustainable development.

5. Regular Maintenance: **Regular maintenance of vehicles and appliances** improves efficiency. **Proper tire pressure** reduces fuel consumption. Efficient machines waste less energy.

6. Use Less, Save More: **Public awareness campaigns** promote energy conservation. **Individual actions** contribute to national energy security. Small habits create large savings when followed by millions.

7. Summary of Chapter 5

Key Points from NCERT

1. Minerals are homogeneous, naturally occurring substances with definite chemical composition and structure. They are **non-renewable resources**. They need **careful conservation**.

2. Types of minerals:

- **Ferrous minerals** contain iron, such as **iron ore** and **manganese**.
- **Non-ferrous minerals** do not contain iron, such as **bauxite, mica, copper, lead, zinc, and gold**.
- **Non-metallic minerals** include **limestone, marble, slate, and granite**.

3. Major minerals:

- **Iron ore** includes **magnetite** and **hematite** and is essential for the steel industry.
- **Bauxite** is the ore of **aluminum**.
- **Mica** is important for **electrical and electronic industries**.
- **Limestone** is the basic raw material for **cement industry**.

4. Mineral conservation is necessary because minerals are **finite and non-renewable**. Methods include **planned mining, improved technology, recycling, substitutes, and waste reduction**.

5. Energy resources are of two types: conventional and non-conventional. Conventional sources include **coal, petroleum, natural gas, and electricity**. Non-conventional sources include **solar, wind, tidal,**

geothermal, biogas, and nuclear energy.

6. Conservation of energy resources is necessary to reduce pollution, save resources, reduce imports, and protect future generations.